

2_10xHuman_8CLC_STARsolo

August 31, 2026

```
[1]: library(Seurat)
library(ggplot2)
library(pheatmap)
library(dplyr)
library(RColorBrewer)
getwd()
dir.create("figures_10xHuman_8CLC")
dir.create("data_10xHuman_8CLC")

colorDict = c("8CLC"="#88535A",
              "ESC"="#F2CC8F")
```

Loading required package: SeuratObject

Loading required package: sp

Attaching package: 'SeuratObject'

The following objects are masked from 'package:base':

intersect, t

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
'/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'
```

```
Warning message in dir.create("figures_10xHuman_8CLC"):
```

```
"'figures_10xHuman_8CLC' already exists"
```

```
Warning message in dir.create("data_10xHuman_8CLC"):
```

```
"'data_10xHuman_8CLC' already exists"
```

```
[2]: dataset_id <- "10xHuman_8CLC"
      sample <- "DUX4"

      path <- paste0("/mnt/volume_1p5T/results/STARsolo_EM_TE/", dataset_id, "/"
        ↪, sample, "/TE_Solo.out/Gene/")

      # get filtered barcodes
      STAR_path <- paste0("/mnt/TEresults/snakemake_results/results/STARoutdir/"
        ↪, dataset_id, "/", sample, "/best_Solo.out/Gene")
      filteredBarcodes <- read.table(paste0(STAR_path, "/filtered/barcodes.tsv"))$V1
        ↪# read barcodes selected by STARsolo

      STARsolo_TE_EM_mat <- Seurat::ReadMtx(
        mtx = paste0(path, "/raw/UniqueAndMult-EM.mtx"),
        cells = paste0(path, "/raw/barcodes.tsv"),
        features = paste0(path, "/raw/features.tsv")
      ) # read matrix
      STARsolo_TE_EM_mat <- STARsolo_TE_EM_mat[, filteredBarcodes] # keep only
        ↪barcodes passing thresholds

      nCells <- length(filteredBarcodes)
      thrMinCells <- round(nCells * 0.02)
      thrMinCells

      objTE_STARsolo <- Seurat::CreateSeuratObject(STARsolo_TE_EM_mat,
        project = "STARsolo_TE_EM",
        min.cells = thrMinCells, min.features = 0
      )

      objTE_STARsolo
```

2

Warning message:

"Feature names cannot have underscores ('_'), replacing with dashes ('-')"

An object of class Seurat

225007 features across 125 samples within 1 assay

Active assay: RNA (225007 features, 0 variable features)

1 layer present: counts

```
[3]: annotation_stellarscope <- read.table("annotation/annotation_stellarscope_hg38.
      ↪tsv") # generated with annotation_scripts/create_annotations_hg38.Rmd
      head(annotation_stellarscope)
```

A data.frame: 6 × 10

	chr	start	end	sw_score	strand	locusID	family	subfamily
	<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>	<chr>
1	chr1	67108754	67109046	1892	+	L1P5	L1	L1P5
2	chr1	8388316	8388618	2582	-	AluY	Alu	AluY
3	chr1	25165804	25166380	4085	+	L1MB5	L1	L1MB5
4	chr1	33554186	33554483	2285	-	AluSc	Alu	AluSc
5	chr1	41942895	41943205	2451	-	AluY_dup1	Alu	AluY
6	chr1	50331337	50332274	1587	+	HAL1	L1	HAL1

```
[4]: # Remove some classes of TEs (already not present in SoloTE)
      classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")
      starsoloTEs <- Features(objTE_STARsolo)

      filteredStarsoloTEs <- ↪
      ↪annotation_stellarscope[annotation_stellarscope$stellarscopeID %in% ↪
      ↪starsoloTEs, ] %>%
      ↪dplyr::filter(!class %in% classesToExclude) %>%
      ↪pull(stellarscopeID)

      print("Percentage of TEs in removed orders:")
      print(length(setdiff(starsoloTEs, filteredStarsoloTEs)) / length(starsoloTEs) * ↪
      ↪100)

      objTE_STARsolo <- objTE_STARsolo[intersect(starsoloTEs, filteredStarsoloTEs), ]

      print("Summary of n counts")
      print(summary(objTE_STARsolo$nCount_RNA))
      print("Summary of n feature")
      print(summary(objTE_STARsolo$nFeature_RNA))

      print("Loci present in the annotation:")
      print(table(rownames(objTE_STARsolo) %in% ↪
      ↪annotation_stellarscope$stellarscopeID))
```

```
[1] "Percentage of TEs in removed orders:"
```

```
[1] 0.07955308
```

```
[1] "Summary of n counts"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	4226	6618	8295	10341	52414

```
[1] "Summary of n feature"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	5030	7458	8583	11035	42206

```
[1] "Loci present in the annotation:"
```

```
TRUE  
224828
```

```
[5]: feature_metadata <- annotation_stellarscope[match(Features(objTE_STARsolo),  
  ↪annotation_stellarscope$stellarscopeID),]  
  
head(feature_metadata)
```

```
A data.frame: 6 × 10
```

	chr	start	end	sw_score	strand	locusID	family	subfamily
	<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>	<chr>
2	chr1	8388316	8388618	2582	-	AluY	Alu	AluY
9	chr1	92274206	92275925	536	+	L2	L2	L2
20	chr1	1048489	1048789	2439	+	AluSp	Alu	AluSp
32	chr1	29359333	29360290	8320	+	LTR5_Hs	ERV	LTR5
43	chr1	49282508	49288376	17048	+	L1PA2	L1	L1PA
66	chr1	114294733	114297115	4561	+	HAL1_dup1	L1	HAL1

```
[6]: options(repr.plot.width=7, repr.plot.height=5)  
  
objTE_STARsolo@meta.data$nCount_TE <- objTE_STARsolo@meta.data$nCount_RNA  
objTE_STARsolo@meta.data$nFeature_TE <- objTE_STARsolo@meta.data$nFeature_RNA  
# Visualize QC metrics as a violin plot  
VlnPlot(objTE_STARsolo, features = c("nCount_TE"), ncol = 1,  
  cols = colorDict, pt.size = 0) + theme(text=element_text(size=17))  
ggsave(paste0("figures_", dataset_id, "/nCountTE_violin_STARsolo.pdf"),  
  ↪device='pdf')  
  
VlnPlot(objTE_STARsolo, features = c("nFeature_TE"), ncol = 1,  
  cols = colorDict, pt.size = 0) + theme(text=element_text(size=17))  
ggsave(paste0("figures_", dataset_id, "/nFeatureTE_violin_STARsolo.pdf"),  
  ↪device='pdf')
```

Warning message:

"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."

Warning message:

"The `slot` argument of `FetchData()` is deprecated as of SeuratObject
5.0.0.

Please use the `layer` argument instead.

The deprecated feature was likely used in the [Seurat](#)
package.

Please report the issue at

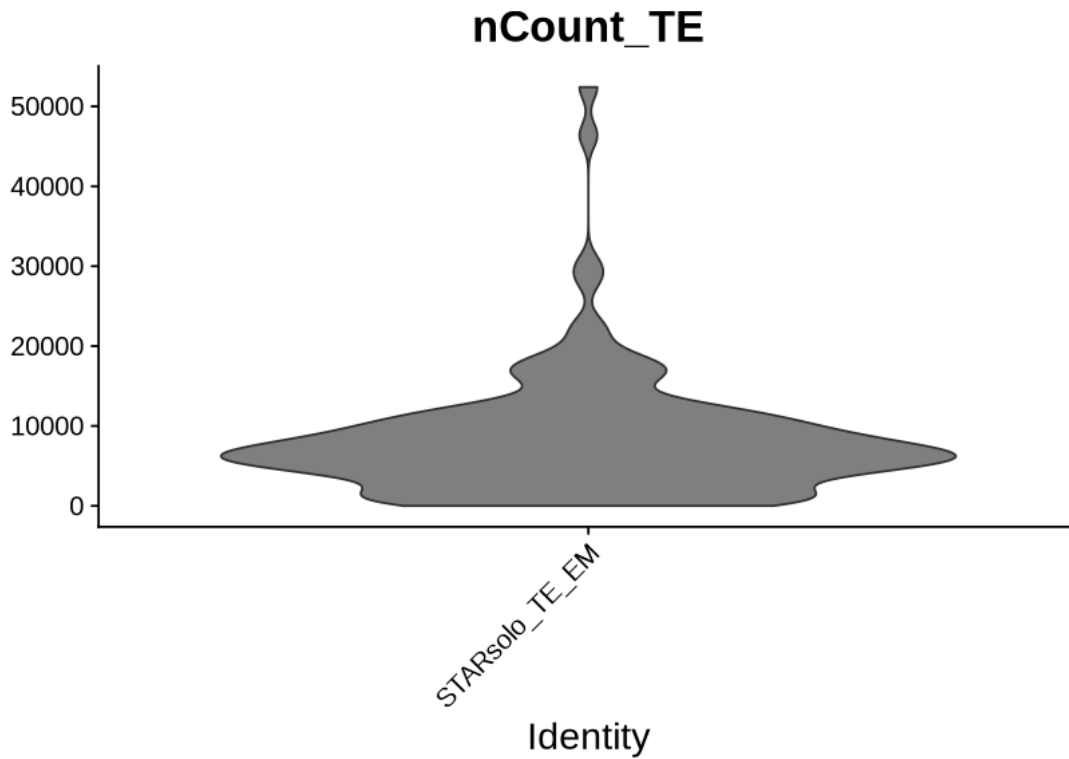
<https://github.com/satijalab/seurat/issues>."

Warning message:

```

"`PackageCheck()` was deprecated in SeuratObject 5.0.0.
Please use `rlang::check_installed()` instead.
The deprecated feature was likely used in the Seurat
package.
Please report the issue at
<https://github.com/satijalab/seurat/issues>."
Warning message:
"No shared levels found between `names(values)` of the manual scale and
the
data's fill values."
Warning message:
"No shared levels found between `names(values)` of the manual scale and
the
data's fill values."
Saving 7 x 7 in image
Warning message:
"No shared levels found between `names(values)` of the manual scale and
the
data's fill values."
Warning message:
"No shared levels found between `names(values)` of the manual scale and
the
data's fill values."
Warning message:
"Default search for "data" layer in "RNA" assay yielded no results; utilizing
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"No shared levels found between `names(values)` of the manual scale and
the
data's fill values."
Warning message:
"No shared levels found between `names(values)` of the manual scale and
the
data's fill values."

```



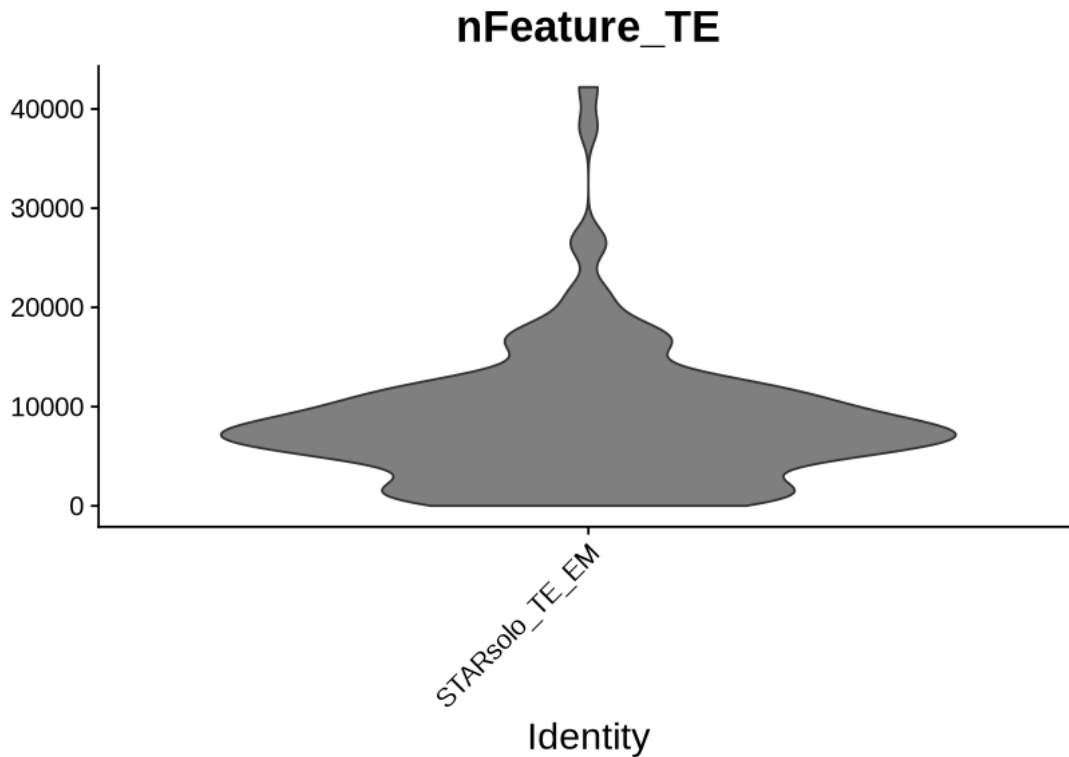
Saving 7 x 7 in image

Warning message:

"No shared levels found between `names(values)` of the manual scale and the data's `fill` values."

Warning message:

"No shared levels found between `names(values)` of the manual scale and the data's `fill` values."



```
[7]: objTE_STARsolo <- JoinLayers(objTE_STARsolo)

objTE_STARsolo <- NormalizeData(objTE_STARsolo, normalization.method = "
  ↪LogNormalize", scale.factor = 10000)

gc()

objTE_STARsolo <- FindVariableFeatures(objTE_STARsolo, selection.method = "
  ↪vst", nfeatures = 4000)

gc()

# Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(objTE_STARsolo), 10)

# plot variable features with and without labels
plot1 <- VariableFeaturePlot(objTE_STARsolo)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot2
```

Normalizing layer: counts

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2 × 6 of type dbl	Ncells	18057352	964.4	48543864	2592.6	43404796	2318.1
	Vcells	111100775	847.7	237901072	1815.1	237781699	1814.2

Finding variable features for layer counts

```
[ ]: gc()
all.genes <- rownames(objTE_STARsolo)
objTE_STARsolo <- ScaleData(objTE_STARsolo) # on hvg
```

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2 × 6 of type dbl	Ncells	18234950	973.9	53878672	2877.5	43931446	2346.2
	Vcells	110883534	846.0	285554796	2178.7	285505392	2178.3

Centering and scaling data matrix

```
[ ]: grep("ERVL", all.genes, value = T)[1:20]
```

1. 'ERVL-B4-int-dup5' 2. 'ERVL-E-int-dup75' 3. 'ERVL-E-int-dup76' 4. 'ERVL-E-int-dup77'
5. 'ERVL-E-int-dup124' 6. 'HERVL40-int-dup12' 7. 'HERVL40-int-dup13' 8. 'ERVL-E-int-dup442'
9. 'ERVL-E-int-dup443' 10. 'ERVL-E-int-dup456' 11. 'ERVL-E-int-dup472' 12. 'ERVL-E-int-dup478'
13. 'ERVL-E-int-dup479' 14. 'ERVL-E-int-dup485' 15. 'ERVL-E-int-dup486'
16. 'ERVL-E-int-dup528' 17. 'HERVL-int-dup142' 18. 'ERVL-E-int-dup604' 19. 'HERVL-int-dup166'
20. 'ERVL-E-int-dup691'

```
[ ]: objTE_STARsolo <- RunPCA(objTE_STARsolo, features = VariableFeatures(object = objTE_STARsolo))

DimPlot(objTE_STARsolo, reduction = "pca") + NoLegend()

ElbowPlot(objTE_STARsolo)
```

PC_ 1

Positive: L1M5-dup2145, MER58B-dup488, L1ME1-dup2966, Charlie11-dup39, L1MC2-dup1546, L1M2c-dup287, L1M5-dup15178, L1ME2z-dup1340, Tigger7-dup678, L1MB4-dup158

L1MB7-dup3353, Plat-L3, L1M4c-dup995, L1MC5a-dup1654, L1MD-dup5329, L1MC1-dup2190, L1MB7-dup2389, L1M3f-dup52, L1MB8-dup3510, L1MD2-dup3074

MER33-dup108, MLT1B-dup1251, L1MA4-dup6243, L1MD2-dup1087, MIRb-dup15873, MIR-dup8731, L1ME1-dup588, L1MB3-dup4934, L1M8-dup32, L2b-dup4964

Negative: L1MCa-dup774, L2-dup4529, MER49, MIR-dup11559, L1MEg-dup156, MLT1H1-dup139, L1M2-dup4884, MLT1A1-dup6104, L1M5-dup586, L1MC5-dup1347

L3-dup2422, L1MC2-dup577, L1MEg-dup3489, L1MB5-dup317, L1M4a1-dup316, HAL1-dup2821, L1MCa-dup778, Tigger8-dup92, MIR3-dup5367, MIR3-dup5366

HAL1-dup848, MER45C-dup248, L1ME3G-dup419, L1M2-dup4320, MIRb-dup15305, L1MA4A-dup93, L1MDa-dup2850, L1M2-dup3636, L1MD3-dup806, L1MA5-dup3973

PC_ 2

Positive: L1MD2-dup1087, Plat-L3, L1MEc-dup94, L1M4c-dup654, L1M3-dup6363, L1MB7-dup958, MLT1B-int-dup14, L1MB4-dup849, L1MB7-dup3353, Tigger7-dup678
L1MB4-dup158, L2-dup672, L1ME1-dup2100, L1ME1-dup1430, L1ME1-dup1448, L1MA9-dup3234, L1MA6-dup2904, L1MC5a-dup883, L1MA8-dup1518, L1M4-dup7121
L2b-dup4964, MIRb-dup15146, MIRb-dup15090, MIRb-dup19712, L1M3f-dup52, L2c-dup4053, L1M1-dup445, L1M4-dup2445, L1M5-dup7998, HAL1-dup1804
Negative: L1M2a-dup192, Charlie1a-dup648, L1M3-dup2888, L3-dup5042, Charlie11-dup39, L1MA4-dup4065, Charlie1-dup771, L1MA5-dup3099, L1MA7-dup1508, L1MC-dup3088

L1M5-dup4660, MLT1D-dup5432, L1MC3-dup3414, MIR-dup35722, MIR-dup18069, L1M2-dup231, L1M5-dup3908, MER58B-dup488, L1MA6-dup2848, MER21B-dup1319

MIR-dup8731, L1M5-dup2145, L1MC4-dup1075, L1MC4-dup1045, L1MC4-dup1011, L1MC4-dup1031, L1MC4-dup1072, L1M4-dup11616, Plat-L3-dup539, L1MA4-dup6966

PC_ 3

Positive: MER21B-dup1485, L1MB3-dup204, MLT1D-dup5088, L1ME3E-dup67, MIR-dup8727, L1ME3-dup492, Tigger17c-dup13, L1M5-dup2633, MLT1E2-dup510, L1MD1-dup375

L1ME4b-dup1085, MLT1H2-int-dup44, L1MC5-dup514, L1MC1-dup1178, L1MCa-dup2392, MLT1F1-dup6, L1MC5-dup2324, L1MEc-dup1101, L1M1-dup1, L1MC1-dup3583

L1M4a2-dup80, L1MA4A-dup110, L1M4-dup1601, L1ME3-dup325, L1M4-dup1644, L1MC5a-dup1430, MER2-dup3104, MLT1F-dup467, L1MA4A-dup2209, L1M4-dup9566

Negative: L1MC1-dup2190, L1M8-dup32, L1M2a-dup192, L1MC-dup3088, L1MA7-dup1508, L1M5-dup4660, MLT1B-dup1251, MER33-dup108, L1ME3-dup1309, L1M4-dup4222

L1M3-dup2888, L1M3-dup5414, L1MD1-dup696, L1MC2-dup1664, L1MC4a-dup1155, L1MD1-dup641, Charlie1-dup771, L1MA4-dup4065, Tigger1-dup873, L1ME1-dup1448

L1ME1-dup1430, L1ME1-dup2100, L1MD2-dup885, L1MD2-dup1084, L1MB2-dup887, L1M5-dup7998, Charlie1a-dup648, MLT2D-dup325, MER44B-dup957, L1MC3-dup3414

PC_ 4

Positive: HAL1M8-dup121, L1ME1-dup2966, L1MC3-dup1030, MIR3-dup6119, L1MB3-dup1265, L1M5-dup2145, MER58B-dup488, L1ME2-dup1523, L1ME2-dup1616, L1MC4-dup4942

MER44A-dup577, L1MB3-dup4115, L2a-dup11252, Charlie11-dup39, L1MC5-dup1662, L1ME2z-dup1340, L1MDa-dup1170, L2a-dup7620, L1MEd-dup2394, L1MC5a-dup1654

L1MA5-dup1711, MER46C-dup53, L1MB8-dup5570, L1MCa-dup2211, MER5A1-dup704, L2a-dup16822, L1MB3-dup5164, MIR-dup404, MIRb-dup28389, MIRc-dup5288

Negative: MER2-dup3104, LTR37-int-dup192, L1M4-dup1930, L1MC4a-dup2481, L1MB8-dup1133, L2c-dup9845, L1MCa-dup811, Tigger1a-Art-dup22, Tigger1-dup505, Tigger1-dup527

Tigger1a-Art-dup20, Tigger1a-Art-dup18, MER3-dup236, MIR-dup35722, MLT1A0-int-dup60, L1MD-dup921, MER5B-dup1937, L1M4-dup4013, MER21B-dup1485, L1M5-dup3954

L1MB3-dup1034, L1MD1-dup213, L1MD1-dup193, L1MEc-dup928, L1MEc-dup939, L1MEc-dup913, MER44A-dup614, L1ME1-dup567, L1MA9-dup87, L1MC5-dup2324
PC_ 5

Positive: MLT1C-dup5722, Tigger1-dup505, Tigger1a-Art-dup20, Tigger1a-Art-dup22, Tigger1a-Art-dup18, Tigger1-dup527, L2a-dup6876, L1MC1-dup4779, L1MD-dup6032, L1MC4-dup5202

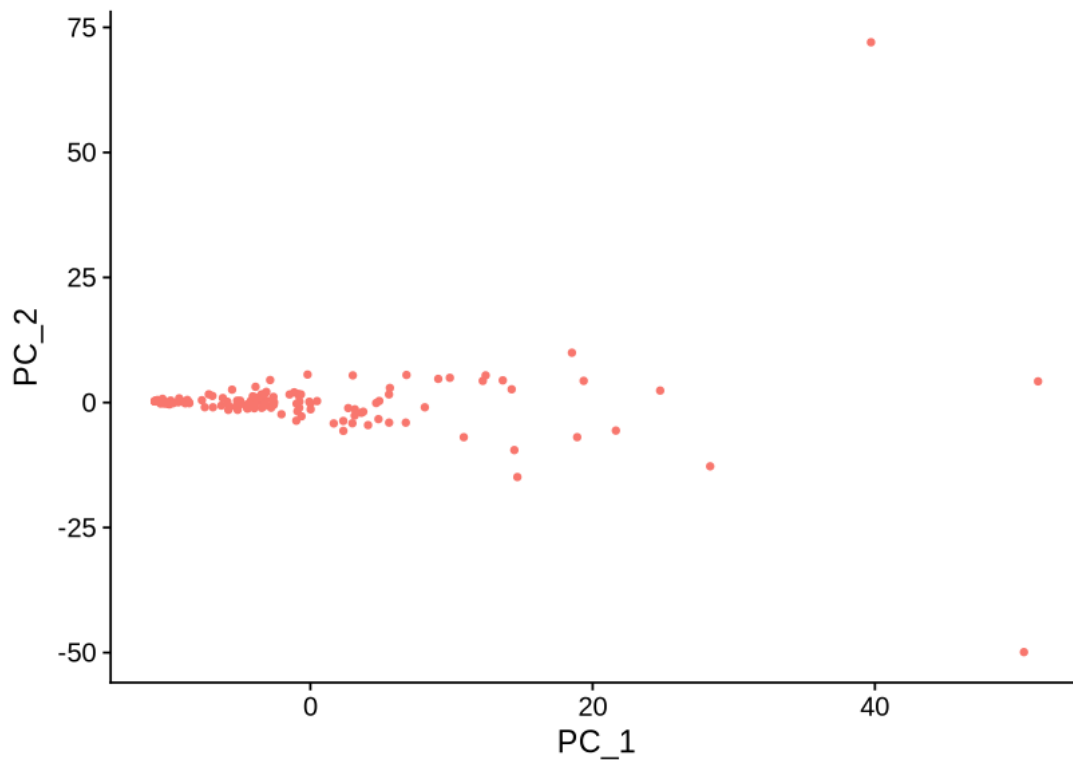
L1MA6-dup4708, L1MC5-dup802, L1MEf-dup732, L1MB8-dup1474, MIRb-dup20486, L1MA5-dup4572, L1MA5-dup3761, L1MB5-dup2937, L1M1-dup257, L1MA4A-dup1285

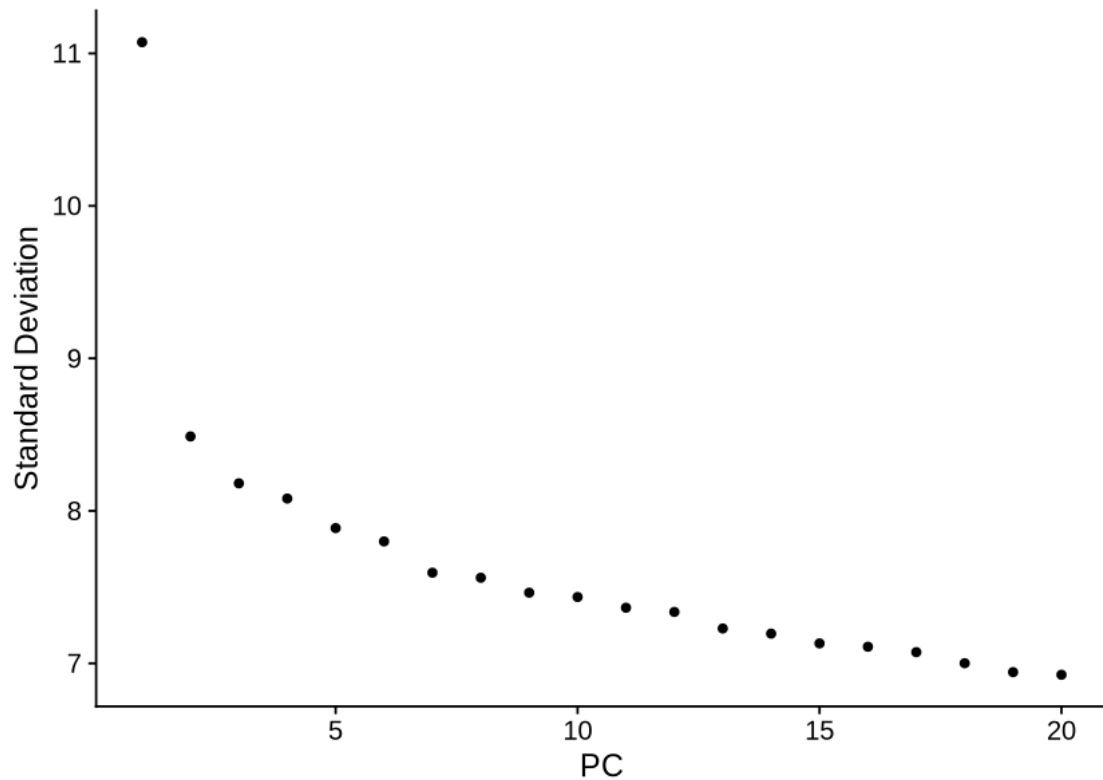
L1MC-dup2385, L1M4-dup2120, L1ME2-dup425, L1M3-dup5949, L1MC5-dup1193, L1MB2-dup2674, L1MEg-dup1824, L2a-dup10442, L1ME1-dup567, L1MB8-dup1757

Negative: L1MC4-dup1072, L1MC4-dup1075, L1MC4-dup1031, L1MC4-dup1011, L1MC4-dup1045, L1MC4-dup966, L1MC4-dup1065, L1MC4-dup1028, L1MC4-dup973, L1MC4-dup1034

L1M2-dup231, L1MC1-dup4485, ERVL-int-dup67, Tigger17c-dup13, HAL1-dup1804, L1MCa-dup884, HAL1-dup385, L2a-dup21058, MLT1B-dup5194, L1MB7-dup4585

L1MEc-dup94, L1ME2-dup1932, MIRb-dup12437, MLT1D-dup1202, L2a-dup11661, L1MB7-dup918, L1MC5a-dup1940, L1MDb-dup108, L1M4c-dup235, L1M5-dup7354





```
[ ]: objTE_STARsolo <- FindNeighbors(objTE_STARsolo, dims = 1:13, k.param = 20)
objTE_STARsolo <- FindClusters(objTE_STARsolo, resolution = 1)
objTE_STARsolo <- RunUMAP(objTE_STARsolo, dims = 1:13)
DimPlot(objTE_STARsolo, reduction = "umap")
```

Computing nearest neighbor graph

Computing SNN

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 125

Number of edges: 4091

Running Louvain algorithm...

Maximum modularity in 10 random starts: 0.4052

Number of communities: 3

Elapsed time: 0 seconds

Warning message:

"The default method for RunUMAP has changed from calling Python UMAP via reticulate to the R-native UWOT using the cosine metric"

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12:15:44 Writing NN index file to temp file
/mnt/TEdataStorage_2T/tmp/RtmpEZ2G4S/file363e1b4202105c

12:15:44 Searching Annoy index using 1 thread, search_k = 3000

12:15:44 Annoy recall = 100%

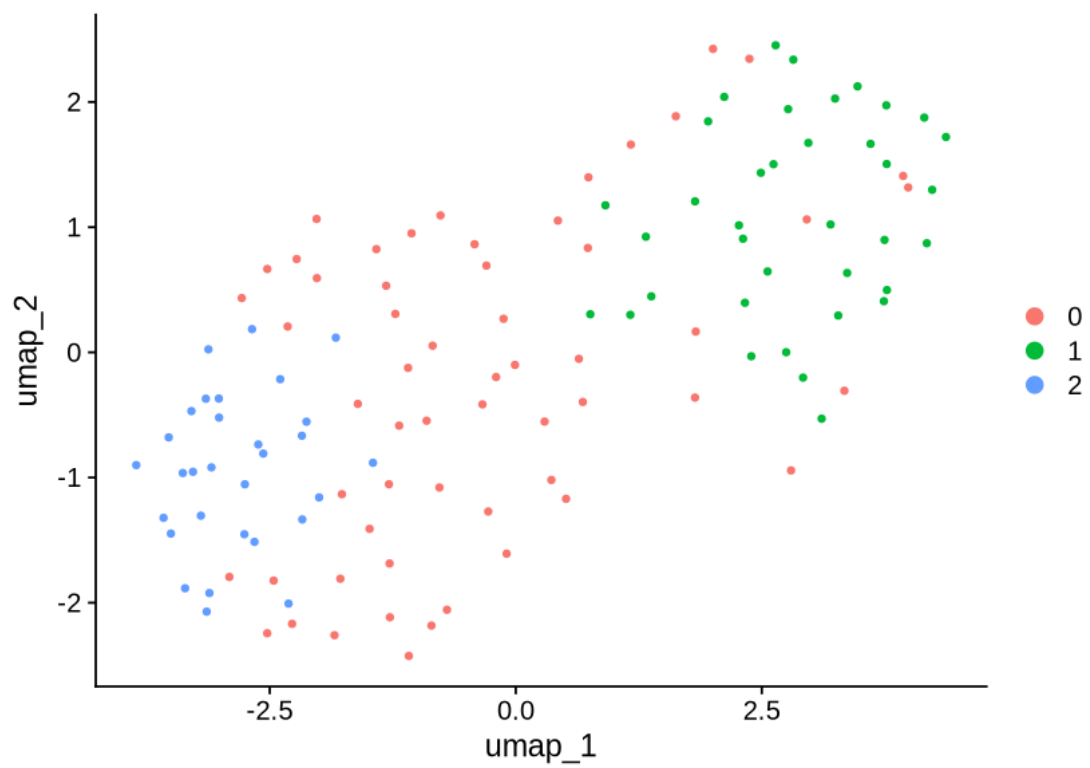
12:15:47 Commencing smooth kNN distance calibration using 1 thread
with target n_neighbors = 30

12:15:52 Initializing from normalized Laplacian + noise (using RSpectra)

12:15:52 Commencing optimization for 500 epochs, with 5106 positive edges

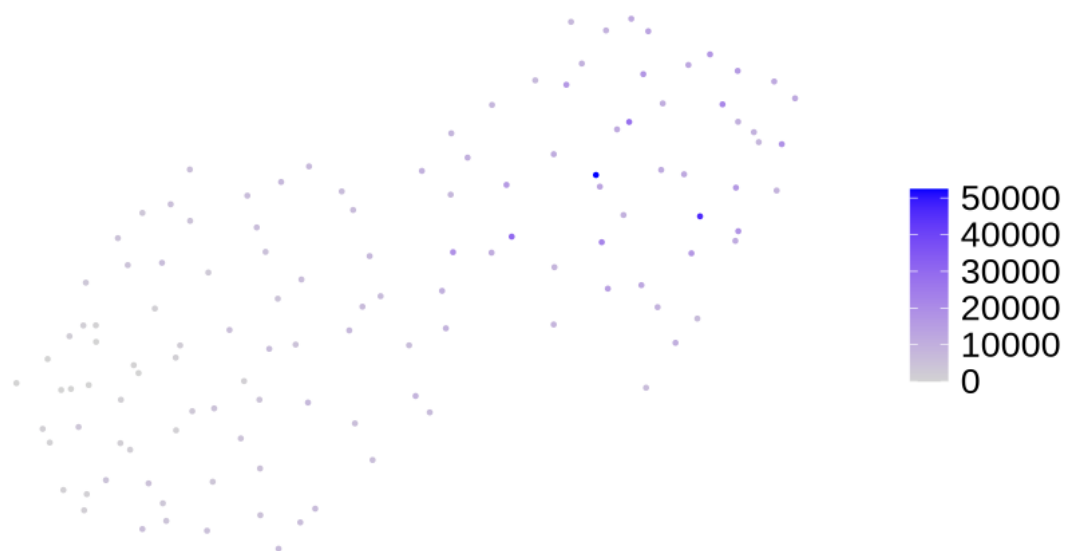
12:15:52 Using rng type: pcg

12:15:55 Optimization finished



```
[ ]: FeaturePlot(objTE_STARsolo, reduction = "umap", features = "nCount_TE", pt.size_
↪ = 0.5) +
  theme_void() +
  theme(text=element_text(size=20))
```

nCount_TE



```
[ ]: saveRDS(objTE_STARsolo, paste0("data_", dataset_id, "/STARsolo_", dataset_id, "_seuratObj.RDS"))
```

```
[ ]:
```